

Project:



Calculations for:	Divn/Dept: Bridges	Job Nr/File No:
Bearing Design in accordance with Freyssinet design manual	Calculated by:	Date:
	Checked by:	Date:

Bearing Design

Movements	
Longitudinal	= 10,60 mm
Transverse	= 4,19 mm

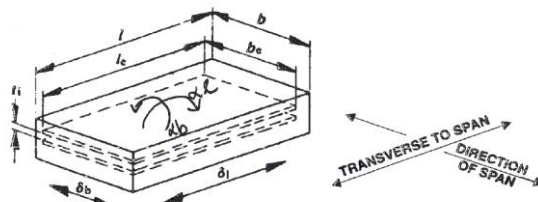
Bearing Size	(457 x 254 x 54)
l	= 457 mm
b	= 254 mm
T	= 54 mm
le	= 447 mm
be	= 244 mm

Plate thk=	4,5 mm
No. of plates=	4

No. of layers=	3
te	= 10 mm
ti	= 10 mm

tq	= 36 mm
Σti	= 30 mm
Ae	= 109068 mm ²
lp	= 1382 mm
G	= 0,9
Vmax=	1420 kN
Vdl	= 384 kN
Vll	= 1036 kN

δb	= 27,83 mm
δl	= 7,64 mm
δr = √(δl ² + δb ²)	28,86 mm
S	= 7,89204



$$\frac{H_s t_q}{A G}$$

Applied forces	
Longitudinal Direction	Horizontal Direction
Hs= 50 kN	Ht= 10 kN
δbH = 17,2 mm	δlH = 3,45 mm

1. Shear Strain	
Eq	= 0,80 FAILS!

1 SHEAR STRAIN

$\epsilon_q = \frac{\delta_r}{t_q}$ must not exceed 0.7
(Freyssinet conservatively use 0.6)

2. Maximum Design Strain	
A1	= 94764,9 mm ²
Ec	= 3,16
Eα	= 0,01
k	= 1,36
Et	= 5,43 FAILS!

2 MAXIMUM TOTAL DESIGN STRAIN

$\epsilon_t = k (\epsilon_c + \epsilon_q + \epsilon_\alpha)$ must not exceed 5
where k = a factor equal to 1.5 for live load effects
1.0 for all other effects

ϵ_c = Nominal strain due to compressive forces and given by the expression $\epsilon_c = \frac{1.5V}{G A_1 S}$

A_1 = Reduced effective plan area due to the loading effects and given by the expression

$$A_1 = A_e \left(1 - \frac{\delta_b}{b_e} - \frac{\delta_l}{l_e} \right)$$

ϵ_q = Shear strain due to translational movements and given by the expression $\epsilon_q = \frac{\delta_r}{t_q}$

ϵ_α = Nominal strain due to angular rotation and given by the expression $\epsilon_\alpha = \frac{(b_e^2 \alpha_b + l_e^2 \alpha_l)}{2 l_i \sum t_i}$

Project:



Calculations for:

Divn/Dept: Bridges

Job Nr/File No:

Calculated by:

Date:

Sheet No

Checked by:

Date:

2 of 3

Bearing Design**3. Reinforcing Plate Thickness****3 REINFORCING PLATE THICKNESS**

To resist induced tensile stresses under load, the minimum thickness of the steel plates should be

$$\frac{1,3V (t_1 + t_2)}{A_1 \sigma_s} \quad \text{but not less than 2mm}$$

$$t_1 = 10 \text{ mm}$$

$$t_2 = 10 \text{ mm}$$

$$\sigma_s = 290 \text{ N/mm}^2$$

 σ_s must not be greater than the yield stress nor greater than 120 N/mm² for plates with holes
290 N/mm² for plates without holes

$$t_{min} = 1,34343 \text{ mm}$$

FAILS, CHECK PLATE SIZE**OK!****4. Stability**

$$V/A_1 = 14,9844 \text{ N/mm}^2$$

4 STABILITY

$$(2 \cdot b_e \cdot G \cdot S') / (3 \sum t_i) = 38,513$$

The mean pressure $\frac{V}{A_1}$ must satisfy the expression $\frac{V}{A_1} < \frac{2b_e G S'}{3 \sum t_i}$

$$b_e/4 = 61 \text{ mm}$$

This expression is satisfied if $\sum t_i < \frac{b_e}{4}$ **OK!****5. Vertical Deflection****5 VERTICAL DEFLECTION**

$$E_b = 2000 \text{ N/mm}^2$$

The total vertical deflection Δ must not exceed the value specified by the engineer and is given by

$$\text{the expression } \Delta = \sum \delta \quad \text{where } \delta = \frac{V \cdot t_i}{5 A_e G S^2} + \frac{V \cdot t_i}{A_e E_b}$$

$$\delta = 0,52961 \text{ mm}$$

(We would recommend that δ should not exceed 15% of t_i).

$$\Delta \text{ Total deflection} = 1,5888 \text{ mm}$$

$$0,15 \cdot t_i = 1,5 \text{ mm}$$

FAILS!**6. Rotational Limit****6 ROTATIONAL LIMITATION**

$$\Delta = 1,59 \text{ mm}$$

The rotation limitation, as specified by the code, is met if the total vertical deflection Δ satisfies the expression $\Delta > \frac{(b_e \alpha_b + l_e \alpha_l)}{3}$

$$(b_e \cdot \alpha_b + l_e \cdot \alpha_l) / 3 = 0,0081$$

OK!

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	Checked by:	Date:	

Bearing Design															
7. Fixing of Bearings															
a) Under all Loading															
H = 83745,5 kN															
0.1(V + 2A ₁) = 160953 kN															
OK!															
b) Under Permanent Loads															
V/A ₁ = 4,05213 N/mm ²															
OK!															
7. FIXING OF BEARINGS If there is insufficient friction to prevent relative movement ("slippage") between the bearing and the structure under the most adverse loading conditions, positive means of location must be provided. Friction may be considered adequate if: a) under all loading conditions, resultant $H < 0,1 (V + 2A_1)$ and b) under permanent loads $\frac{V}{A_1} > 2$ where all terms and their units are as follows: H is the design force exerted by the bearing to resist translational movement (in N) V is the vertical design load effect (in N) A₁ is the reduced effective plan area as defined in step 2 above (in mm²) Supplementary Note: The force exerted on the structure by the bearing resisting translational movement is given by the expression $H = AG \frac{\delta_r}{t_q}$															
BEARING FAILS, THEREFORE CHOOSE NEW BEARING SIZE															